Faculty of Health and Medical Sciences

Day 4 (part 2): Adherence, Intention to treat & Per protocol analyses, estimand framework and Statistical Analysis Plan

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November 12, 2026

Outline/Intended Learning Outcomes (ILOs)

Adherence, Intention to treat & Per protocol analyses
- ILO: exemplify the challenges caused by nonadherence
- ILO: restate and contrast different approaches to analyze data with nonadherence
- ILO: explain their strengths and limitations and how these differ between superiority and non-inferiority studies

Estimand framework
- ILO: restate and exemplify the benefit of using the estimand framework
- ILO: restate and exemplify the core attributes of estimands
- ILO: recognize the challenges posed by intercurrent events and discuss different strategies to handle them
- ILO: explain, using examples, how the estimand framework can improve our understanding of and approach to challenges related to nonadherence and missing data.

Statistical Analysis Plan (SAP)
- ILO: restate the rationale for SAPs and their content
- ILO: recall examples of SAPs and where to find resources SAPs

Nonadherence

> **Nonadherence** in a randomized clinical trial (RCT) occurs when study participants do not follow the randomly assigned treatment protocol. Reasons for nonadherence may include the study participant not taking trial medications, crossing over to the other intervention being studied, or accessing treatment outside of the trial. Nonadherence also may occur when the clinician is unable to complete the assigned therapy (eg, a surgical procedure) as intended.

Source: Smith et al. "Interpreting the results of intention-to-treat, per-protocol, and as-treated analyses of clinical trials." JAMA 326.5 (2021): 433-434.

- Nonadherence is often difficult to avoid.
- Nonadherence should be documented in the clinical study report.

  "The frequency and type of protocol violations, missing values, and other problems should be documented in the clinical study report and their potential influence on the trial results should be described (see ICH E3)." [1]

[1] EMA scientific guidelines: *ICH Topic E9 Statistical Principles for Clinical Trials*, 1998, https://www.ema.europa.eu/en/documents/scientific-guideline/ich-e-9-statistical-principles-clinical-trials-step-5_en.pdf

Analysis sets & analytical approaches

"Intention-to-treat (ITT), per- protocol (PP), and as-treated (AT) approaches to analysis differ in how the included patient population and treatment assignments are defined, with important implications for interpretation of treatment effects in clinical trials."

- *"With the ITT approach, all randomized patients are included in the analysis, based on the groups to which they were initially randomly assigned."*
- *"Per-protocol only analyzes data from participants who follow the protocol, excluding their data after they become nonadherent."*
- *"AT analyses consider the treatment actually received by the participant, without regard to adherence to their randomization assignment."*

Source: Smith et al. "Interpreting the results of intention-to-treat, per-protocol, and as-treated analyses of clinical trials." JAMA 326.5 (2021): 433-434.

Intention-to-treat (ITT) principle

"The intention-to-treat (see Glossary) principle implies that the primary analysis should include all randomised subjects. Compliance with this principle would necessitate complete follow-up of all randomised subjects for study outcomes. In practice this ideal may be difficult to achieve, for reasons to be described. In this document the term 'full analysis set' is used to describe the analysis set which is as complete as possible and as close as possible to the intention-to-treat ideal of including all randomised subjects. Preservation of the initial randomisation in analysis is important in preventing bias and in providing a secure foundation for statistical tests. In many clinical trials the use of the full analysis set provides a conservative strategy." [2]

[2] EMA scientific guidelines: *ICH Topic E 9 Statistical Principles for Clinical Trials*, 1998, https://www.ema.europa.eu/en/documents/scientific-guideline/ich-e-9-statistical-principles-clinical-trials-step-5_en.pdf

Appendix: ITT principle, other quotes

Box 6. Intention-to-treat analysis

The special strength of the RCT is the avoidance of bias when allocating interventions to trial participants (see Box 1). That strength allows strong inferences about cause and effect that are not justified with other study designs. In order to preserve fully the huge benefit of randomisation we should include all randomised participants in the analysis, all retained in the group to which they were allocated. Those two conditions define an "intention-to-treat" analysis, which is widely recommended as the preferred analysis strategy [18,223]. Intention-to-treat analysis corresponds to analysing the groups exactly as randomised. Strict intention-to-treat analysis is often hard to achieve for two main reasons—missing outcomes for some participants and non-adherence to the trial protocol.

Source: Moher et al (2010). CONSORT 2010 Explanation and Elaboration: updated guidelines for reporting parallel group randomised trials. Journal of Clinical Epidemiology, 63(8), e1–e37

Principle 4.2. Intention-to-treat (ITT) *The primary comparison in a clinical trial should be of treatments to which patients were randomized, not of treatments actually received.*

Another way to phrase the ITT principle is that one should analyze according to how one intended to treat the patients regardless of whether they received treatment, hence the name, intention-to-treat.

Remark 4.3. *Two important consequences of the ITT principle are:*

1. *One should never stop following patients who do not comply with their randomized treatment.*
2. *Missing data threaten the integrity of a clinical trial by making it difficult to compare the original randomized groups.*

The ITT principle is a tough pill to swallow, so to speak. For instance, the ITT principle

Source: page 36, in *Statistical Thinking in Clinical Trials*, by Michael Proschan (2022)

Why is ITT (often, but not always!) conservative?

- *"We start by considering placebo-controlled double-blind RCTs. It is well known that if treatment A has a null effect on the outcome, then both the effect of assigned treatment Z and the effect of treatment A will be null."*
- *"This is a key advantage of the ITT analysis: it correctly estimates the effect of treatment A under the null, regardless of the adherence pattern."*
- *"It is also well known that if treatment A has a non-null effect (that is, either increases or decreases the risk of the outcome) and some participants do not adhere to their assigned treatment, then the effect of assigned treatment Z will be closer to the null that the actual effect of treatment A."*
- *"This bias toward the null is due to contamination of the treatment groups: some subjects assigned to treatment (Z=1) may not take it (A=0) whereas some subjects assigned to placebo (Z=0) may find a way to take treatment (A=1). As long as the proportion of patients who end up taking treatment (A=1) is greater in the group assigned to treatment (Z=1) than in the group assigned to placebo (Z=0), the effect of assigned treatment Z will be in between the effect of treatment A and the null value."*

Source: Hernán & Hernández-Díaz. *Beyond the intention-to-treat in comparative effectiveness research.* Clinical trials 9.1 (2012): 48-55

Example & Exercise.4.1. use R!

In Exercises 1.1 and 1.2 (Day 1), we have seen that we could include 1048 patients to have a power of 80%. The investigator expected 11% and 17% because they expected 40% of the patients to be in septic shock, with risk 30% vs 20%; and 60% not in septic shock, with risk 7.5% vs 5%. Let's assume that, in fact, the above expectations were under perfect adherence. But unfortunately, the investigator suspected that up to 10% of the patients in septic shock randomized to the new treatment will likely end up not receiving it, for unfortunate logistical reasons. Let's assume that for these patients, the risk is the same as in the control group (30%) and that for those who receive it the risk is the same as under perfect adherence (20%). Further assume that we plan to perform an ITT analysis.

Questions:

1. What is the risk we can expect to estimate, in average, in the experimental treatment arm, with an ITT analysis (instead of 11%)?
2. What does the power become, if we indeed expect this risk instead of 11% (versus 17%)?
3. By how much could we inflate the sample size to compensate for nonadherence and still have 80% power?
4. Remember, we expect to be able to include 130 patients per year. By how long should we prolong the accrual period, and consequently, the trial duration?

When is ITT analysis **NOT** conservative? Why? (1/2)

ITT is not conservative in non-inferiority trials.

- Contamination will usually make the two arms more similar than they should. Hence, an ITT analysis will produce bias towards "no difference between the two arms" and a treatment effect biased towards zero. In non-inferiority trials, this makes it is "easier" to conclude that the treatment effect is below the non-inferiority margin; and it facilitates concluding non-inferiority. This increases the risk of falsely concluding non-inferiority (i.e., the risk of type-I error might be inflated).[3]

Hence, ITT analyses should be complemented.

- *"In a superiority trial the full analysis set, based on the ITT (intention-to-treat) principle, is the analysis set of choice, with appropriate support provided by the PP (per protocol) analysis set. In a non-inferiority trial, the full analysis set and the PP analysis set have equal importance and their use should lead to similar conclusions for a robust interpretation."* [4]

[3] See e.g., Scenario 1 in Hernán & Hernández-Díaz. Clinical trials 9.1 (2012): 48-55.

[4] EMA scientific guidelines: *Points to Consider on switching between superiority and non-inferiority*, 200, https://www.ema.europa.eu/en/documents/scientific-guideline/points-consider-switching-between-superiority-and-non-inferiority_en.pdf

Exercise.4.2: read and practice statistical thinking!

Now let us consider double-blind RCTs that compare two active treatments. These trials are often designed to show that a new treatment ($A=1$) is not inferior to a reference treatment ($A=0$) in terms of either benefits or harms. An example of a noninferiority trial would be one that compares the reduction in blood glucose between a new inhaled insulin and regular injectable insulin. The protocol of the trial would specify a noninferiority margin, that is, the maximum average difference in blood glucose that is considered equivalent (e.g., 10 mg/dL). Using an ITT comparison, the new insulin ($A=1$) will be declared not inferior to classical insulin ($A=0$) if the average reduction in blood glucose in the group assigned to the new treatment ($Z=1$) is within 10 mg/dL of the average reduction in blood glucose in the group assigned to the reference treatment ($Z=0$) plus/minus random variability. Such ITT analysis may be

Source: Hernán & Hernández-Díaz. Clinical trials 9.1 (2012): 48-55.

Scenario 1

The new treatment $A=1$ is actually inferior to the reference treatment $A=0$, for example, the average reduction in blood glucose is 10 mg/dL under treatment $A=1$ and 22 mg/dL under treatment $A=0$. The type and magnitude of adherence is equal in the two groups, for example 30% of subjects in each group decided not to take insulin. As a result, the average reduction is, say, 7 mg/dL in the group assigned to the new treatment ($Z=1$) and 15 mg/dL in the group assigned to the reference treatment ($Z=0$). An ITT analysis, which is biased toward the null in this scenario, may incorrectly suggest that the new treatment $A=1$ is not inferior to the reference treatment $A=0$.

Questions:

1. Why 7 and 15 mg/dL?
2. What are the expected values of the mean difference in reduction under perfect adherence versus non-adherence, in this example, when using an ITT analysis?
3. What is the expected sign of the $E(Z)$ in each case? What can we conclude on the type-I error? Help: type-I error $= 2\Phi\{E(Z) - 1.96\}$ for a usual two-sided test at the desired 5% level and $\Phi(-1.96) = 2.5\%$, with $\Phi(z) = P(Z \leq z)$ when Z is a standard normal random variable.
4. (Advanced) Assume that the trial was planned to have a power of 80% under the assumption that the treatment were actually equally efficacious. Can we compute the type-I error of the ITT analysis under this assumption? Help: $E(Z) = 2.8$ under the alternative for which the trial is powered, $E(Z) = (\delta - \delta_0)/SE$ and $SE = \sigma\sqrt{2/n}$. Also, if treatments are equally efficacious, then $\delta = 0$.

When is ITT NOT conservative? Why? (2/2)

Other (rare/specific) cases exist:

"Finally, if neither treatment has any effect on the outcome, the ITT is generally considered unbiased. However, there is the potential for bias away from the null compared with the average causal effect of treatment when 1) a treatment is beneficial for some participants and harmful for others, and 2) the proportion of adherence is related to the outcome, such as when some of the side effects causing nonadherence are also markers for the outcome." [5]

[5] Shrier et al. *The intention-to-treat analysis is not always the conservative approach.* The American journal of medicine 130.7 (2017): 867-871.

ITT analyses can be very conservative

3. An excessive conservatism. Imagine a double-blind crossover placebo-controlled study of a new drug for the prophylaxis of angina. Fig. 2 shows both the placebo and active drug data on one hypothetical patient. Here the intended dose of the new drug was one per day, and if the patient had taken the new drug as intended, there would be no reason to think it was superior to placebo on the basis of the similar frequency of anginal attacks on both treatments.

But what if a medication monitor had been used,[4] and the pattern of dosage was the one listed as *Actual* in Fig. 2? It seems the patient did not take any drug at all for the first 2 days and had anginal attacks on days 2 and 3. He then took the drug for 3 days without angina, stopped taking the drug, and had an attack the next day. The same thing happened again after 1 week of full compliance without angina. Surely, if there were multiple records like this one and the placebo records did not show similar patterns, we would be justified in concluding that the drug is effective, would we not? Apparently not: Many statisticians would tell us that we must do someting called an "intention to treat" analysis. In this case such an analysis

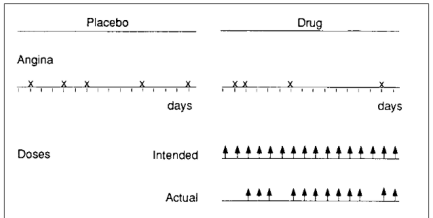


Fig. 2. Hypothetical angina prophylaxis data on one patient.

Source: Sheiner, L.B. (1991). The intellectual health of clinical drug evaluation. Clinical Pharmacology and Therapeutics 50 (1): 4–9

ITT analysis: effectiveness instead of efficacy?

Definitions:
"Effectiveness is usually defined as 'how well a treatment works in everyday practice,' and efficacy as 'how well a treatment works under perfect adherence and highly controlled conditions." [6]

Does ITT analysis evaluate effectiveness instead of efficacy? Not necessarily.

- *"Arguments in favor of the ITT principle include [...] (2) analysis by randomized groups irrespective of compliance tells what will happen in the real world if we attempt to treat patients* [i.e., it estimates effectiveness] [..] *Opponents of the ITT principle counter the second point above with the argument that compliance during the trial and outside the trial are two different things. If the trial shows that the intervention works, that will likely lead to better compliance in the real world."* [7]
- *"The result of an intention-to-treat analysis is affected by the trial-specific pattern of adherence to the treatment strategies under study and therefore may not be directly relevant for guiding decisions in clinical settings with different adherence patterns. In fact, the publication and dissemination of the intention-to-treat result may change adherence in clinical settings, possibly rendering the result outdated."*[8]

[6] Hernán & Hernández-Díaz. *Beyond the intention-to-treat in comparative effectiveness research.* Clinical trials 9.1 (2012): 48-55.
[7] page 36, in *Statistical Thinking in Clinical Trials*, by Michael Proschan (2022)
[8] Hernán & M. Robins. *Per-protocol analyses of pragmatic trials.* N Engl J Med 377.14 (2017): 1391-1398.

Per-Protocol analysis: what and why?

- What?
  "Study participants who do not take the assigned therapy are excluded in the analyses and may also be censored from the analyses when treatment is discontinued. PP analyses can also be defined where study participants who do not adhere to protocol (e.g., schedule of assigned therapy) or have protocol violations (e.g., took prohibited medications or the wrong therapy) might be excluded from the analyses. [...] *The exact definition of the PP analysis set is not unique and frequently varies depending on the goals of the analyses.*
- Why?
  The PP analyses may be viewed as an evaluation of the treatment under near ideal use of the therapy (e.g., good adherence), and thus may be informative when adherence can be influenced by clinician intervention. A PP analyses can intuitively be an attractive alternative as it is more consistent with the theory that "treatment not taken cannot affect the outcome" than ITT." [9]
- Common recommendation:
  "In general, it is advantageous to demonstrate a lack of sensitivity of the principal trial results to alternative choices of the set of subjects analysed. In confirmatory trials it is usually appropriate to plan to conduct both an analysis of the full analysis set and a per protocol analysis, so that any differences between them can be the subject of explicit discussion and interpretation." [10]

[9] Source: page 2028, in *Fundamental Concepts for New Clinical Trialists*, Scott Evans & Naitee Ting (2015).
[10] EMA scientific guidelines: *ICH Topic E 9 Statistical Principles for Clinical Trials*, 1998, https://www.ema.europa.eu/en/documents/scientific-guideline/ich-e-9-statistical-principles-clinical-trials-step-5_en.pdf

PP and AT analyses: limitations

Randomization is broken, biases can arise:
"As-treated and PP analyses are not simple to interpret because of the potential loss of an important benefit of randomization: the elimination of systematic bias in treatment assignment. Selection bias [in PP analysis] *and confounding* [in a AT analysis] *in the treatment effect estimate arises if patients who are more adherent with the assigned treatment differ in ways that also influence their outcomes compared with those who are less adherent."*[11]

Adjustment for confounding might be needed:
"In summary, 'as treated' and 'per protocol' analyses transform RCTs into observational studies for all practical purposes. The estimates from these analyses can only be interpreted as the effect of treatment A if the analysis is appropriately adjusted for the confounders L. If the intended analysis of the RCT is 'as treated' or 'per protocol,' then the protocol of the trial should describe the potential confounders and how they will be measured, just like the protocol of an observational study would do." [12]

- Simple baseline covariate adjustments might be sufficient in some case, but not always.
- Sensitivity analyses may be needed.
- Advanced methods to deal with time-dependent confounding might be necessary. In that case, to properly adjust for confounding *"it will be necessary to collect and analyze postrandomization data on adherence to the assigned treatment strategies and on the evolution of prognostic factors."*[13]

[11] Smith et al. JAMA 326.5 (2021): 433-434.
[12] Hernán & Hernández-Díaz. Clinical trials 9.1 (2012): 48-55.
[13] Hernán & M. Robins. *Per-protocol analyses of pragmatic trials.* N Engl J Med 377.14 (2017): 1391-1398.

Important take-home messages

(I) Different approaches to data analysis will often result in estimates of different treatment effects because they correspond to different definitions of the treatment effect being studied.

(II) Should we choose a default analysis method and accept the treatment effect it implicitly defines and estimates? Not necessarily.

(III) Should we instead first define the treatment effect that is most relevant to the scientific question and then choose a data analysis that is aligned with that definition? Probably.

(IV) In practice, do many analyses fall somewhere in between these two extremes? Yes. In many settings, there is a trade-off between relevance and estimability: the treatment effect of greatest scientific interest is often difficult to estimate reliably from the available data.

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Estimand framework
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Hence the need for estimands...

... to clarify what is the goal of the data analysis and, consequently, the right interpretation of the results.

Received: 25 July 2019 | Revised: 11 October 2019 | Accepted: 20 October 2019
DOI: 10.1111/bcp.14195

EMA GUIDELINES

Beyond "Intent-to-treat" and "Per protocol": Improving assessment of treatment effects in clinical trials through the specification of an estimand

Surprisingly, even a careful examination of the trial protocol or of the methods section in a scientific publication is often insufficient to find fundamental information: what exact research question does the trial seek to answer? This can be articulated into several subquestions that are not answered explicitly in many trial protocols or scientific publications: what is the medical outcome of interest and when and how is this measured for each individual patient? To which patients should the trial results exactly apply: to all those who could have entered the trial or all those who could tolerate the treatment long enough? Which effect is of interest in a setting where patients can, eg, switch to other treatments or have access to rescue therapy? Is it the effect of starting the new treatment, irrespective of subsequent treatment strategies? Is the study designed and is data collection planned based on these choices (see Section A.4. Impact on Trial Design and Conduct)? Stating the targeted analysis set or principle, eg intention to treat or per protocol, does not fully capture these choices and their consequences for design, data collection and analysis (see Section A.5. Impact on trial analysis). The way trial objectives are usu-

Source: Pétavy et al. British Journal of Clinical Pharmacology 86.7 (2020): 1235-1239.



Missing data: another common problem (partly solved by the estimand framework)

The need for estimand also came from recurrent misunderstandings and lack of clarity about:

- what missing data in clinical trials really are
- how they should be analyzed (Which statistical methods? simple? advanced?)
- And why! I.e., which treatment effect do we wish to estimate, when there are missing data? (it often relates to the cause of missingness)

Historical note:
Actually, the concept of "estimand" was popularized by an important report of the National Research Council entitled *"The Prevention and Treatment of Missing Data in Clinical Trials"* (2012)[14], written by an impressive panel of experts, following a request from the U.S. Food and Drug Administration (FDA) in 2008.

[14] Freely available at: https://doi.org/10.17226/12955



A famous case (FDA vs Astra-Zeneca & Bristol-Myers Squibb, 1/2)

An example where the definition of the treatment effect left room for interpretation was discussed in 2011 in a public advisory committee meeting. Astra-Zeneca and Bristol-Myers Squibb presented the data on their anti-diabetic drug dapagliflozin (FDA 2011). The primary endpoint for their randomized controlled pivotal trials was the change in glycated hemoglobin (HbA1c) from baseline to 24 weeks. For ethical reasons, patients were allowed to receive glucose-lowering rescue medications, if one of several markers of glycemic control exceeded prespecified thresholds. The statistical analyses included all randomized patients who received at least one dose and for whom the HbA1c baseline value and at least one post-baseline value were available. Measurements collected after initiation of rescue medication were set to missing. This appears to be common practice in the analysis of diabetes trials as intake of rescue medication may mask differences in glycemic values between treatment groups, especially as it will occur more frequently in less effective treatment groups. Following the tradition at the time, the last prerescue-medication value was carried forward and analyzed as the end-of-trial value.

The statistical reviewer at the FDA commented on the primary analysis using the last observation carried forward (LOCF) approach as follows (FDA 2011):

> "While FDA has implicitly endorsed LOCF imputation for diabetes trials in the past, there is now more awareness in the statistical community of the limitations of this approach. […]
>
> My own preferred analysis simply uses the observed values of patients who were rescued. This approach may seem counterintuitive if one believes that rescue treatment makes the subsequent outcomes less relevant to evaluation of the test agent. It has the virtue, however, of respecting the intent-to-treat principle, in the sense that the analysis is based on the randomized treatment rather than the treatment actually received (i.e., planned treatment plus rescue)."

At first glance it seems that both parties have simply used a different statistical approach to analyze the data. On the one hand, the sponsors removed the data after initiation of rescue medication and applied the LOCF approach to impute the resulting missing data. On the other hand, the FDA statistician refrains from using LOCF due to the poor statistical properties and includes all data regardless of initiation of rescue medication.

Akacha et al (2017), *Estimands and Their Role in Clinical Trials*, Statistics in Biopharmaceutical Research, 9:3, 268-271.



A famous case (FDA vs Astra-Zeneca & Bristol-Myers Squibb, 2/2)

> But the implication of using a different statistical approach to analyze the data goes far beyond statistics. The implied scientific questions of interest that are answered by the two parties are fundamentally different. The sponsors seem to be interested in establishing the treatment effect of the initially randomized treatments had no patient received rescue medication, while the FDA statistician is comparing the treatment policies "dapagliflozin plus rescue" versus "control plus rescue."

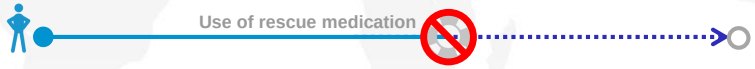
> As shown in this example, differences in the scientific question may arise due to different ways of handling data after trial treatment discontinuation or intake of rescue medication, although this ambiguity may not be clear to the clinicians or statisticians involved in the trial, neither on the sponsor nor on the regulatory side. The current practice of defining trial objectives rather vaguely has led to various challenges in interactions with different stakeholders, including regulators and payers. To improve transparency and ensure alignment between trials objectives and statistical approaches, it appears necessary to go back to first principles and to clearly define what is to be estimated. Once there is agreement on the estimand of interest, or in other words "that which is to be estimated," this estimand will further inform appropriate trial design, data capture, and analysis.

Akacha et al (2017), *Estimands and Their Role in Clinical Trials*, Statistics in Biopharmaceutical Research, 9:3, 268-271.

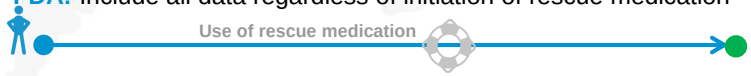
Source for next slides: E9(R1) Working group training materials, `https://www.ich.org/page/efficacy-guidelines`, latest accessed 27/10/2026.

Example: Type II diabetes mellitus

- **Different perspectives on the inclusion of data:**
  - **Applicant:** Remove data after initiation of rescue medication, and impute.



  - **FDA:** Include all data regardless of initiation of rescue medication



- In this case, absence of an agreed treatment effect of interest led to a primary analysis that was not aligned to the treatment effect of interest preferred by the regulator.
  - **ITT principle …** *"followed up, assessed and analysed as members of that group irrespective of their compliance to the planned course of treatment"* **requires a comparison of treatment policies** 'dapagliflozin plus rescue' versus 'control plus rescue'.

ITT: Intention-to-treat

Example: Type II diabetes mellitus

- The applicant had not specified the treatment effect of interest:



It should not be the choices for data collection and data analysis that determine the question of interest!

- The choice of trial design and data collection might have resulted in difficulties for estimation or inadequate statistical power for the analysis, and differences in interpretation of the trial data

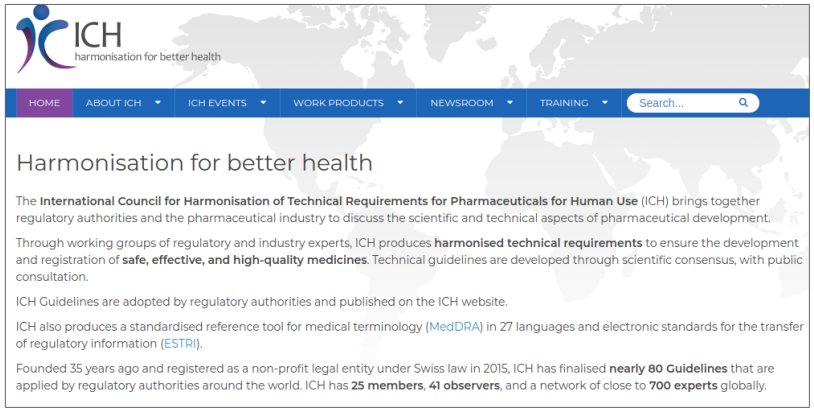
This illustrates the importance of reaching agreement over what to estimate: **the estimand.**

Estimand

Estimand A precise description of the treatment effect reflecting the clinical question posed by the trial objective. It summarises at a population-level what the outcomes would be in the same patients under different treatment conditions being compared.

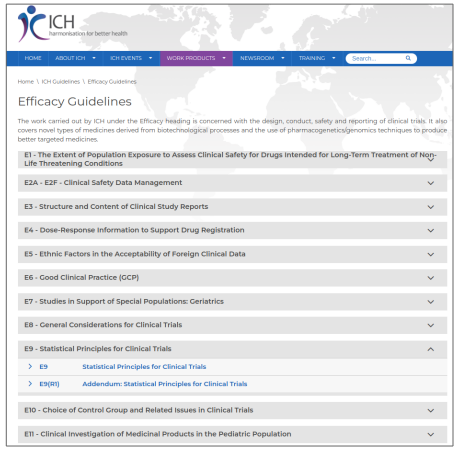
Estimand definition, ICH E9(R1) addendum

Digression: ICH (1/3)



`https://www.ich.org/`

Digression: ICH (2/3)



- Shown E1 to E11 only, but E1 to E23 exists (as of 2026/05/29)

`https://www.ich.org/page/efficacy-guidelines`

Digression: ICH (3/3) (Who draft these guidelines? example for E20)





Note: non-exhaustive list of members of the working group; see `https://www.ich.org/page/efficacy-guidelines`

The estimand framework (1/2)

- **What?**
  "estimands describe the treatment effect the study sets out to quantify for a given outcome. They do so using a structured approach, with standardised terminology."
- **Why?**
  "The structured approach ensures that all aspects of the treatment effect are described, while the use of standardised terminology ensures that the estimand can be easily understood."
- **How?**
  "Table 3 lists the five core attributes that comprise an estimand: population, treatment conditions, endpoint, summary measure, and the strategies used to handle each type of intercurrent event in the treatment effect definition."

Table 3 | Core attributes of estimands

Attribute	Definition
Population	Patients for whom researchers want to estimate the treatment effect
Treatment conditions	Different intervention strategies being compared in the treatment effect definition
Endpoint	Outcome for each participant that is used in the treatment effect definition
Summary measure	Method used to summarise and compare the endpoint between treatment conditions (eg, risk ratio, odds ratio)
Handling of intercurrent events	Strategies used to handle each intercurrent event* in the treatment effect definition; different strategies could be used for different types of intercurrent events

*Intercurrent events are post-baseline events (or post-randomisation events in randomised trials) that affect the interpretation or existence of outcome data. These events frequently affect receipt of treatment (eg, treatment switching or treatment discontinuation) or preclude existence of the outcome (eg, death, if it is not defined as part of the outcome).

Source: Kahan, Brennan C., et al. "The estimands framework: a primer on the ICH E9 (R1) addendum." BMJ 384 (2024).

The estimand framework (2/2)

"Estimands are rapidly becoming an essential part of medical research. They serve to remind us of this important truth: the research question should drive our study methods, and not the other way around."

[...]

"Estimands extend the commonly used PICO (population, intervention, comparator, outcome) framework for defining research questions by adding two additional attributes: the summary measure, which defines how outcomes are summarised and compared between treatments; and the strategies used to handle each type of intercurrent event, which define how things such as treatment switching or treatment discontinuation are handled in the treatment effect definition."[15]

[15] **Source:** Kahan, Brennan C., et al. "The estimands framework: a primer on the ICH E9 (R1) addendum." BMJ 384 (2024).

Example: Type II diabetes mellitus

- In this case FDA and Applicant use different approaches to deal with the **use of rescue medication**.

  This is an *"intercurrent event"*.

- We can identify two different treatment effects:
  - "to assess the treatment effect **if rescue medication is not available**";
  - "to assess the treatment effect **regardless of whether rescue medication is used**".

- These are appropriately thought of as **different questions (estimands)** and not two different answers to the same question.

Intercurrent events

Intercurrent events Events occurring after treatment initiation that affect either the interpretation or the existence of the measurements associated with the clinical question of interest. It is necessary to address intercurrent events when describing the clinical question of interest in order to precisely define the treatment effect that is to be estimated.

Intercurrent events definition, ICH E9(R1) addendum



- These events (e.g., death) that preclude the observation of the variable, should **not be confused** with missing data resulting from **loss to follow-up**!
- **Discontinuation of randomised treatment** represents an intercurrent event whilst **study withdrawal** gives rise to missing data to be addressed in the statistical analysis.

Intercurrent events

Intercurrent events: Events occurring after treatment initiation that affect either the interpretation or the existence of the measurements associated with the clinical question of interest. It is necessary to address intercurrent events when describing the clinical question of interest in order to precisely define the treatment effect that is to be estimated.

Intercurrent events definition, ICH E9(R1) addendum



Unlike missing data, intercurrent events are not to be thought of as a drawback to be avoided in clinical trials, as they can also occur in clinical practice!

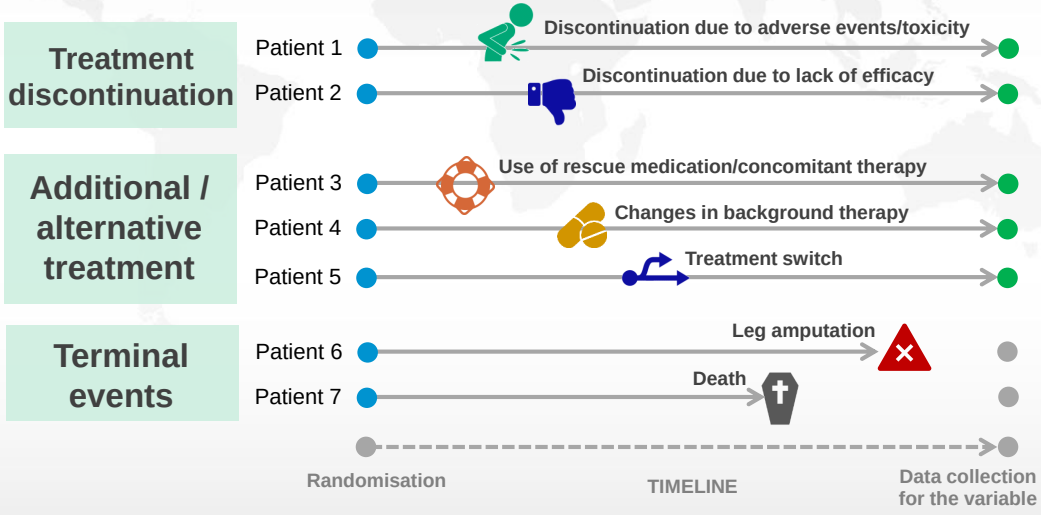
Check addendum section A.3.2!

Strategies to address intercurrent events

A "**strategy**" reflects the choice made on **how to address intercurrent events**, in order to describe the **treatment effect** that is targeted.

- The addendum introduces **five strategies** that can be used alone or in combination to **address different intercurrent events.**
  - Treatment policy
  - Hypothetical
  - Composite
  - While on treatment
  - Principal stratum

Intercurrent events - examples



Strategies to address intercurrent events

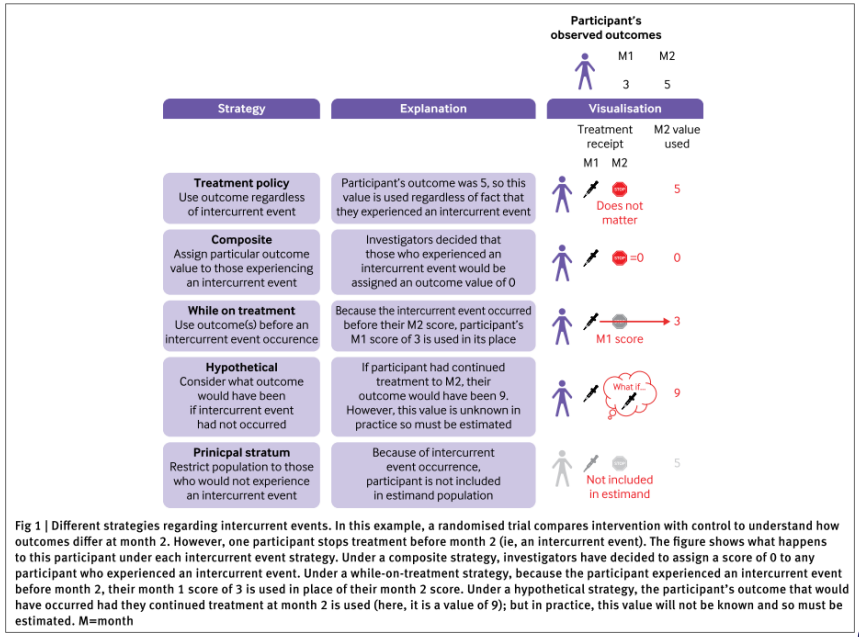


Fig 1 | Different strategies regarding intercurrent events. In this example, a randomised trial compares intervention with control to understand how outcomes differ at month 2. However, one participant stops treatment before month 2 (ie, an intercurrent event). The figure shows what happens to this participant under each intercurrent event strategy. Under a composite strategy, investigators have decided to assign a score of 0 to any participant who experienced an intercurrent event. Under a while-on-treatment strategy, because the participant experienced an intercurrent event before month 2, their month 1 score of 3 is used in place of their month 2 score. Under a hypothetical strategy, the participant's outcome that would have occurred had they continued treatment at month 2 is used (here, it is a value of 9); but in practice, this value will not be known and so must be estimated. M=month

Source: Kahan, Brennan C., et al. "The estimands framework: a primer on the ICH E9 (R1) addendum." BMJ 384 (2024).

Strategies and modeling assumptions (1/3)

- Often, the "treatment policy" and "composite" strategies let the data speak freely. They do not require strong modeling assumptions for the statistical analysis to provide reliable results.

- By contrast, the "hypothetical" and "principal stratum" strategies typically require strong, untestable, modeling assumptions. This is a serious limitation.[16]

  Using covariate adjustments and regression models (e.g., linear or logistic models), we typically try to identify subgroups of patients who are sufficiently similar to each other to "guess" what the outcome and/or intercurrent event would have been in counterfactual settings, using data from similar patients for which the outcome and/or intercurrent event was observed (e.g., if intercurrent events had not happened and/or if the patient had been randomized to, and received, the alternative treatment).

  Using sensitivity analyses, we can (and should!) try to assess the robustness of the results to different modeling assumptions.

[16] see e.g., Fleming et al. *"A perspective on the appropriate implementation of ICH E9 (R1) addendum strategies for handling intercurrent events."* Statistics in Medicine 44.10-12 (2025): e70104.

Strategies and modeling assumptions (3/3)

"We argue that for an estimand to be useful, it must be both relevant and reliable (Box 1), that is, it must address a question that matters to stakeholders (relevance) and enable investigators to answer the question with a high degree of confidence (reliability) [14, 19]. Estimands that lack either relevance or reliability can hamper decision making by stakeholders. For estimands that lack relevance, there is no guarantee that any observed benefit will translate to a benefit that matters to stakeholders; for estimands that lack reliability, stakeholders cannot be certain that any observed benefit is real. In both cases, uncertainty may remain around the intervention's utility, regardless of the actual results."

BOX 1 | What makes an estimand useful for decision making?

- Estimands will be able to usefully inform decision making if they are both *relevant* and *reliable*.
- An estimand will be *relevant* if it answers a question that matters to stakeholders.
- If an estimand lacks relevance, then any benefits or harms observed for that estimand may not translate to benefits or harms on estimands that *do* matter to stakeholders.
- An estimand will be *reliable* if it can be estimated with a high degree of confidence, i.e., can be estimated under plausible assumptions that are likely to be true.
- If an estimand lacks reliability, stakeholders cannot be sure that any observed benefits or harms are real and not just due to inherent bias in the estimator.
- In some trials, there may not be an estimand which is both perfectly relevant and perfectly reliable.
- In these trials, there will necessarily be a trade-off between relevance and reliability, i.e., investigators may need to choose an estimand that is "good enough" on one of the domains.
- However, investigators and stakeholders should strive to ensure any estimand has sufficient relevance and reliability to adequately inform decision making.

Source: Kahan et al, *What Makes an Estimand Useful? Guidance on the Choice of Intercurrent Event Strategies*, Statistics in Medicine 45, no. 6–7 (2026), https://doi.org/10.1002/sim.70452.

Strategies and modeling assumptions (3/3)

Sensitivity analysis Is a series of analyses conducted with the intent to explore the robustness of inferences from the main estimator to deviations from its underlying modelling assumptions and limitations in the data.

Sensitivity analysis definition, ICH E9(R1) addendum

Complicated & technical topic (but in short):

- use different regression model assumptions (e.g., linear or non-linear trends)
- use different structural assumptions (e.g., different set of covariates to define patients who are sufficiently similar to each other)
- use best case / worst case scenarios, or other in between, to impute (missing) counterfactual outcomes (e.g., what if all, none, or 50% those who have received rescue medication had survived/died, if they had not received it?)

Exercise 4.3: practice statistical thinking!

ILO: learn from examples and practice arguing for or against the use of specific strategies.

1. Read all 8 examples in Table 1, provided at the next three slides.
2. Prepare yourself for a short oral presentation of one example (assigned to you) in group of $\approx$ 4 students.
   - 2.1 Describe the setting and the suggested strategy to handle the intercurrent event(s).
   - 2.2 Restate the strengths and limitations highlighted by the authors.
   - 2.3 What is your view about those? Do you agree fully, partly, or not at all? Why? What further information would you need to reach a more confident conclusion?

TABLE 1 Examples where strategies other than treatment policy (or composite for mortality) may be preferable for decision making.

Strategy	Setting	Challenge with treatment policy, or composite for mortality	Why alternative strategies may be preferable
Principal stratum			
Example 1: Safety outcomes in double-blind placebo-controlled trials (e.g., comparing daily tablets vs. placebo tablets)	Safety outcomes such as adverse events and toxicities are essential to decision making. It is often of interest to understand the harms directly attributable to the assigned treatment; however, this becomes challenging when some participants do not receive any dose of study medication.	A treatment policy strategy attributes any harms to the assigned treatment, even if the participant did not receive any dose of study medication. This can affect interpretation of actual safety signals seen in the trial.	A principal stratum strategy addresses the effect in patients who would receive at least one dose of study medication (either intervention or placebo) regardless of their treatment assignment; thus, harms are only attributed to a treatment if the participant received any dose of that treatment, which may be more relevant than treatment policy. In this setting, a principal stratum strategy can be robustly estimated by using a modified intention-to-treat approach which restricts the analysis to those who received at least one dose of study medication; because of the double-blinding, the participants who do not begin treatment will, by definition, be comparable in each treatment arm and randomization is preserved among the modified intention-to-treat population [9].
Example 2: Comparison of different interventions during surgery	Many trials compare different anesthetic approaches to manage patients during surgery, or different techniques of delivering the surgery itself. Because these interventions may require advanced warning to prepare, participants are often randomized before their surgery begins. Some participants who are randomized may subsequently have their surgery canceled, e.g., due to deteriorating health.	A treatment policy strategy compares different anesthetic approaches/surgical techniques whether or not patients undergo surgery. However, these interventions cannot be used in patients who don't undergo surgery, and so this comparison lacks relevance.	A principal stratum strategy addresses the effect of different anesthetic approaches/surgical techniques in those who would always undergo surgery regardless of treatment assignment, and thus actually require the intervention – a clearly more relevant approach than treatment policy. If the decision to cancel or undergo surgery is not affected by treatment allocation—an assumption that can be justified by contextual knowledge in some trials (e.g., see references 15 and 19)—then this strategy can also be robustly estimated by using a modified intention-to-treat analysis which restricts the analysis population to those who actually undergo surgery [9].
Hypothetical			
Example 1: Trials affected by treatment interruptions due to COVID-19 lockdowns or other extenuating circumstances, such as wars, natural disasters, etc.	At several points during the COVID-19 pandemic, the United Kingdom and other countries entered a lockdown. During this time many trials experienced treatment interruptions, where participants were not able to access or receive study treatment in the same way as they could outside of the pandemic lockdown.	These treatment interruptions caused by COVID-19 lockdowns (or other extenuating circumstances such as wars, natural disasters, etc.) are unlikely to reflect standard use of the intervention(s) going forward. As such, a treatment policy strategy is unlikely to provide an accurate reflection of the intervention's effect in real-world (post-pandemic) circumstances.	A hypothetical strategy can be used to address the intervention's effect in real-world (post-pandemic) circumstances, in which treatment interruptions due to lockdowns would not occur. Because the mechanism behind the intercurrent event will be known for many trials (i.e., investigators can identify deterministically which participants were affected and why), a hypothetical strategy could be estimated robustly using assumptions known to be valid (e.g., such that a certain subset of participants had treatment discontinued due to the lockdown, rather than any individual characteristics) [10, 11].

(Continues)

TABLE 1 (Continued)

Strategy	Setting	Challenge with treatment policy, or composite for mortality	Why alternative strategies may be preferable
Example 2: Non-inferiority trials with trial-specific intercurrent events	Trial-specific intercurrent events are those that occur in the trial setting but would not occur outside of it, in the context the interventions would be used [12]. Examples include non-adherence due to (i) external disruptions such as COVID-19; (ii) clinical uncertainty around the intervention's safety which would be resolved once trial data is available; or (iii) study conduct issues (e.g., in a high-turnover environment, some study staff not being aware of the trial) [12]. Such trial-specific intercurrent events can make the trial treatment arms artificially similar [12], and in some contexts may be difficult or impossible to fully prevent through the study's design.[a]	A treatment policy strategy would ignore the artificial similarity between treatment arms caused by trial-specific intercurrent events, and thus increase the risk of a spurious non-inferiority finding [12].	A hypothetical strategy addresses the treatment effect in the absence of trial-specific intercurrent events, which is of clear relevance to stakeholders as it (a) addresses the artificial similarity between treatment arms caused by the trial-specific intercurrent events; and (b) matches the likely setting the interventions would be used in. However, because a hypothetical strategy cannot generally be reliably estimated in these contexts owing to strong, untestable assumptions (with some exceptions), for many trials a hypothetical strategy should be used alongside a treatment policy strategy on the basis that a non-inferiority finding for both offers stronger evidence than either alone that the results are not due to trial-specific events (see Morgan et al. for a full discussion [12]).
While-alive			
Example 1: Clinical symptoms in severely ill populations	Many trials conducted in severely ill populations at high risk of death (such as in palliative care) do not try to prolong survival, but instead try to ameliorate clinical symptoms to improve patients' quality of life during the remainder of their life. Mortality during the study is expected to be high, but in many trials should not differ between treatment arms, as the interventions are not intended to affect this. However, in-study mortality affects the existence of such clinical symptom outcomes and must be addressed as part of the estimand.	A composite strategy, in which patients who die are assigned a poor outcome value, does not accurately reflect the objective of these trials, which is to improve patients' quality of life while they are alive. This can affect interpretation of results, and potentially reduce the ability to detect genuine benefits on outcomes of interest [13].	A while-alive strategy addresses the impact of the intervention on their symptoms (i.e., how they feel and function) while they are alive, an issue of direct relevance to patients and clinicians [14]. When the intervention does not affect mortality, as would be the case in certain settings, the while-alive can be reliably estimated and thus offer robust evidence for decision makers [14].
Example 2: Health-related quality of life (HRQoL) in trials where the intervention is expected to have a modest survival benefit but with increased toxicity	Many trials, particularly in settings like oncology, evaluate interventions which aim to provide a modest increase in survival, but may have increased toxicity. Understanding this trade-off is essential for decision making, meaning that outcomes such as HRQoL play a key role for stakeholders.	A composite strategy may mask any drawbacks of the intervention on HRQoL, meaning that stakeholders cannot properly consider the relative trade-offs of the intervention [14]. For instance, consider a new intervention which improves median overall survival by around 3 month, but reduces HRQoL by around 15 points (on a 0–100 scale); a composite strategy could (a) reduce this amount such that it may not appear clinically important; and (b) increase variability so that a statistically significant drop in HRQoL would be unlikely to be found [13].	A while-alive strategy addresses the effect of the intervention on HRQoL while patients are alive, which enables stakeholders to properly consider trade-offs between increased survival and likely reductions in HRQoL [14]. Because survival differs between treatment arms, a while-alive strategy means that the outcome may be measured across different time-points for some patients, which can cause issues around interpretation. However, despite these drawbacks, a while-alive strategy arguably still provides a clearer picture of the relative benefits and harms of the intervention than a composite strategy [14].

TABLE 1 (Continued)

Strategy	Setting	Challenge with treatment policy, or composite for mortality	Why alternative strategies may be preferable
Composite			
Example 1: Use of rescue medication where rescue medication comes with major drawbacks	In many trials, rescue medication is used to ameliorate symptoms. For example, levodopa can be used to control symptoms in Parkinson's disease, but it becomes ineffective over time. Higher levodopa use in one treatment arm may make that treatment appear to be more effective than the other. However, levodopa comes with some drawbacks, including: (i) it can have long-term side effects; and (ii) because it wears off, early use of levodopa may mean worse symptom control later in the patient's life.	A treatment policy strategy may allow the treatment condition which receives more rescue medication to appear to be as or more effective than the one which receives less; however, this effectiveness may be driven primarily by the rescue medication, rather than the treatment itself. However, when the rescue medication has long-term risks or side effects, these may not be adequately captured by the treatment policy strategy, and even supplementary analyses which compare rates of rescue medication use between treatment arms may be insufficient to fully inform decision makers.	A composite strategy (e.g., using an outcome such as "symptom control at 1 year without the use of levodopa") may better enable stakeholders to assess the short-term benefits of the intervention without sacrificing potential long-term benefits (e.g., fewer side-effects and the ability to manage symptoms at a later point using levodopa). This approach can be reliably estimated, and thus offers a potentially relevant and robust estimand for decision making.
Example 2: Non-trial interventions for clinical events	Many trials aim to prevent clinical events. For example, trials of different platelet transfusion strategies in patients with low platelet counts may seek to determine which strategy is best at preventing major bleeding. For some bleeds, patients may receive non-trial interventions, such as blood transfusions. Receipt of such interventions may be included as part of the outcome definition using a composite strategy (e.g., "major bleed requiring blood transfusion").	Use of a treatment policy strategy (i.e., ignoring the occurrence of the blood transfusion) would require the outcome to be defined without reference to the transfusion. This may have drawbacks, including difficulty in finding a clear way to differentiate major from minor bleeding, which may ultimately lead to minor bleeding events being classified as major, leading to challenges around interpretation of study results.	A composite strategy facilitates a clear distinction between minor and major clinical events, which can make both the measurement of the outcome easier, as well as clarify interpretation of results for stakeholders. This approach can be reliably estimated, and thus offers a relevant and robust estimate for decision making.

[a]While ideally trial-specific events should be prevented by design, in practice they may still occur despite strong design principles, either because they cannot be prevented by design (e.g., external disruptions such as COVID-19), or because no amount of training/oversight process will fully eradicate all sources of protocol deviations.

Source: Kahan et al, *What Makes an Estimand Useful? Guidance on the Choice of Intercurrent Event Strategies*, Statistics in Medicine 45, no. 6–7 (2026), https://doi.org/10.1002/sim.70452.

A recent example (Danish investigator-initiated RCT, completed in 2026)

Is hydrochloric acid (HALT) in central venous catheter effective to treat Bloodstream Infections, among children and adults with cancer or hematologic disorders?



(investigator-initiated trial)

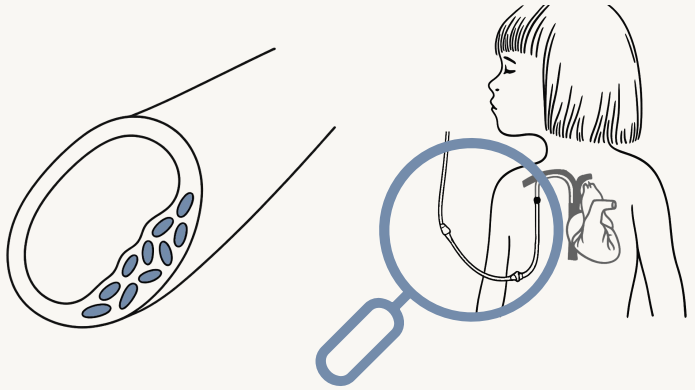
- **ClinicalTrials.gov ID:** NCT05376566
- **Link to the statistical analysis plan:** https://cdn.clinicaltrials.gov/large-docs/66/NCT05376566/SAP_000.pdf
- **Acknowledgments:** this slide and the followings are from previous work by Mette Bondo Mønster & Ulrikka Nygaard.

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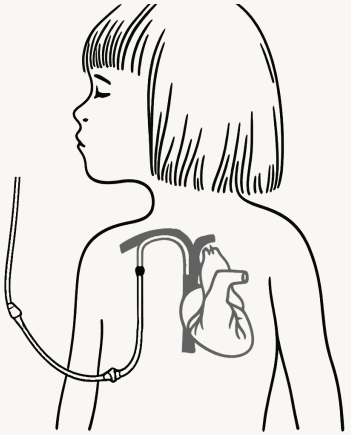
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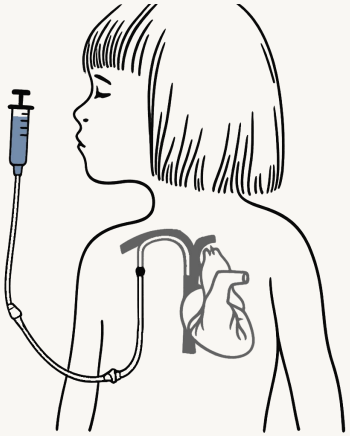
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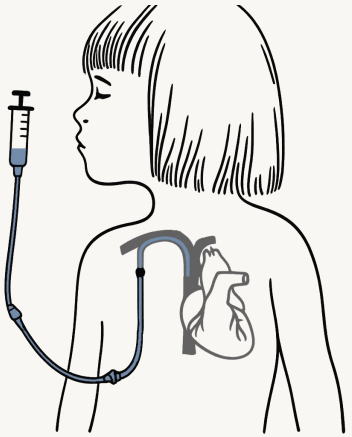
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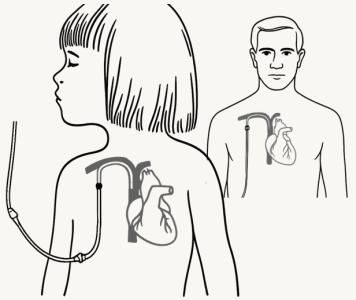
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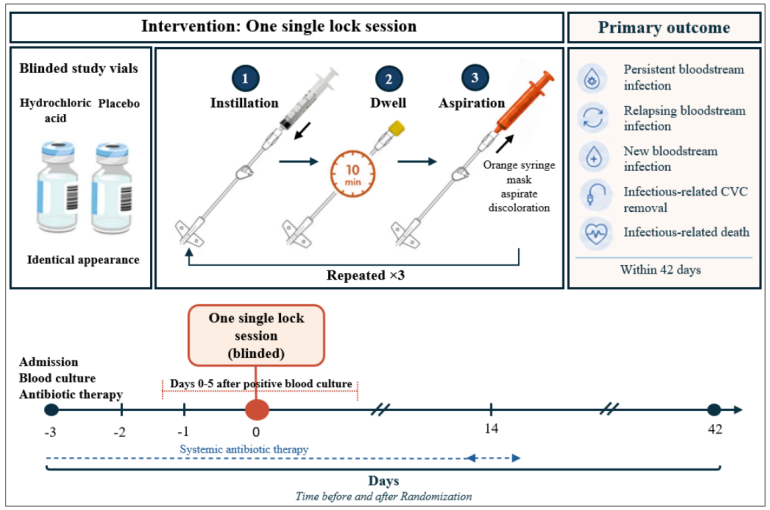
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Trial design



Exercise 4.4: HALT RCT, which estimand? why?

1. Read pages 6-9 of the statistical analysis plan available:
   `https://cdn.clinicaltrials.gov/large-docs/66/NCT05376566/SAP_000.pdf`
2. Discuss the two estimands in small groups ($\approx$ 4 students). Especially:
   - 2.1 Explain to each other what they mean,
   - 2.2 And the rationale for considering both.
   - 2.3 List the important intercurrent events,
   - 2.4 and the different strategies to handle them,
   - 2.5 and critically discuss their rationale (strengths / limitations)
   - 2.6 The second component of the primary outcome can be decomposed in two:
     - new CLABSI with a different bacteria isolate
     - new CLABSI with same bacterial isolate

     Can you think of pros and cons of including the first? Think about the consequence for the interpreation of the results from different perspectives (e.g., patients or researchers) and for the power of the trial.
3. Then we will summarize altogether.

Digression: explanatory versus pragmatic trials

- The estimand framework relates to earlier discussions about explanatory versus pragmatic trials.

1. Definition of the Treatments

1.1. "Equalized" or "optimal" conditions

Consider a trial of anti-cancer treatments in which radiotherapy alone is to be compared with radiotherapy preceded by the administration of a drug which has no effect by itself but which may sensitise the patient to the effects of radiation. Suppose the drug is to be administered over a 30-day period. The "radiotherapy alone" group may then be handled in two different ways (Fig. 1):

(a) radiotherapy may be preceded by a blank period of 30 days, so that it is instituted at the same time in each group;
(b) radiotherapy may be instituted at once, thereby carrying it out at what is most probably the optimal time.

Neither procedure can be said to be "better" than the other. The first allows us to compare two groups which are alike from the radiotherapy point of view and which differ solely in the presence or absence of the drug. It therefore provides an assessment of the sensitising effect of the drug and gives valuable information at a biological level. The second procedure enables us to compare two treatments under the conditions in which they would be applied in practice. We distinguish the two procedures as stemming from two different approaches to the trial, the first *explanatory*, the second *pragmatic*.

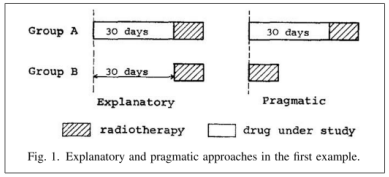


Fig. 1. Explanatory and pragmatic approaches in the first example.

Source: Schwartz & Lellouch (1967). Explanatory and pragmatic attitudes in therapeutic trials. Journal of Chronic Diseases 20: 637–648.

Digression: are estimands old wine in a new bottle?

"Estimands are both new and old, and missing data as well as intercurrent events in clinical trials are a long existing issue in medical research. Conceptually, it appears the estimands are as "old" as medical research and clinical trials, because it always had estimand elements (e.g. outcome measured) and even if empirically estimated, there was a target of estimation. Estimands in the shape principled by ICH E9(R1) are an innovative solution to deal with fundamental elements of clinical trials, starting from the research question and dealing with intercurrent events, missing data and treatment effect definitions. The estimand framework provides a new framework to align key elements of design, conduct and analysis of clinical trials to adequately answer the clinical question at hand."[17]

[17] Mitroiu, Oude Rengerink, Teerenstra, Pétavy and Roes. *A narrative review of estimands in drug development and regulatory evaluation: old wine in new barrels?*. Trials 21, 671 (2020)

Still new...

"The centrality of estimands to clinical trials is currently not reflected in methods recommended by the Cochrane group or the CONSORT statement, the current standard for reporting clinical trials in medical journals. We encourage revisions to these guidelines."[18]

- To the best of my knowledge, this is still the case.
- The estimand framework is still new. Most of us are still learning how to best use it and write about it in protocols, SAPs and papers.



[18] Keene et al. BMC medicine 21.1 (2023): 276.

Estimand vs Estimator (recap)

- Estimand framework is about what we would like to estimate.
- It is not about how to estimate what we would like (at least not directly).
- For some estimand choices, it might be challenging to come up with a valid statistical analysis (unbiased) that do not rely on questionable assumptions.
- Unmeasured confounding and/or questionable non-testable modeling or structural assumptions might be needed.
- Sensitivity analyses are very much encouraged when questionable assumptions are used.
- Some statistical methods for handling missing data can be used with some estimands. Interesting literature reviews about methods to handle missing data in clinical trials exist.[19]



[19] See e.g., National Research Council. 2010. *The Prevention and Treatment of Missing Data in Clinical Trials*. Washington, DC: The National Academies Press. https://doi.org/10.17226/12955

Outline/Intended Learning Outcomes (ILOs)

Adherence, Intention to treat & Per protocol analyses
- ILO: exemplify the challenges caused by nonadherence
- ILO: restate and contrast different approaches to analyze data with nonadherence
- ILO: explain their strengths and limitations and how these differ between superiority and non-inferiority studies

Estimand framework
- ILO: restate and exemplify the benefit of using the estimand framework
- ILO: restate and exemplify the core attributes of estimands
- ILO: recognize the challenges posed by intercurrent events and discuss different strategies to handle them
- ILO: explain, using examples, how the estimand framework can improve our understanding of and approach to challenges related to nonadherence and missing data.

Statistical Analysis Plan (SAP)
- ILO: restate the rationale for SAPs and their content
- ILO: recall examples of SAPs and where to find resources SAPs



Statistical Analysis Plan (SAP): What and Why?

What is it?

Statistical Analysis Plan

A statistical analysis plan is a document that contains a more technical and detailed elaboration of the principal features of the analysis described in the protocol, and includes detailed procedures for executing the statistical analysis of the primary and secondary variables and other data.

EMA scientific guidelines: *ICH Topic E9 Statistical Principles for Clinical Trials*, 1998, https://www.ema.europa.eu/en/documents/scientific-guideline/ich-e-9-statistical-principles-clinical-trials-step-5_en.pdf

Why writing a SAP?

"The SAP is finalized after the trial protocol is finalized and before the blind is broken. A primary rationale for the SAP is to provide transparency. The SAP clearly prespecifies the planned analyses (endpoints, populations, time points, and analytical strategy) thus providing multiplicity context and assisting with control of the Type I error rate. The SAP further clarifies that any unspecified analyses are post hoc. This helps to limit issues of revisiting the data and iterative analyses that can create multiplicity concerns (although some post hoc analyses inevitably occur due to questions that arise from the data). The SAP also serves as a pseudo-contract between statisticians and other members of the project team helping to preserve statistical resources by preventing numerous iterations of analyses."[20]



[20] Page 201 in *Fundamental Concepts for New Clinical Trialists*, by Scott Evans & Naitee Ting (2015).

Statistical Analysis Plan (SAP): How?

How to write it?

- *"Readers should not have to infer what was probably done; they should be told explicitly."*[21] The higher the level of evidence you seek and the more detailed/rigorous the SAP should be.
- Precise definitions of outcomes (especially primary and key secondary), analysis sets and statistical methods used are essential to describe. Writing about estimands is increasingly promoted.
- Prespecification matters mostly for confirmatory analyses. Consequently, being transparent and precise about the primary and secondary outcomes analyses is essential. It is less important to write about e.g., standard descriptive analyses (although it does not hurt and often helps).
- It helps to look at SAPs from others (often available from the supplementary material of publications) and guidelines/tutorials (see next slides).



[21] Altman DG. Better reporting of randomised controlled trials: the CONSORT statement. BMJ 1996; 313: 570–71

Resources for SAP writing

- TransCelerate template: extensively used in the pharmaceutical industry. Maybe too detailed for most academic trials, but provides good inspiration, especially about how to write about estimands.

  https://www.transceleratebiopharmainc.com/assets/clinical-content-reuse-solutions/ (Accessed: 2025-10-11)

- Template from Stevens et al:[22] focused on academic trials, provides many examples to include in suggested SAP sections, but nothing about estimands.
- Guidelines from Gamble et al:[23] (mostly) focused on academic trials. Easy to read, valuable, rather consensual, but nothing about estimands.
- Chapter in Evans & Ting's book:[24] clear, easy to read, provides example of a SAP outline and rationale. Refer to ICH guidelines and provides historical remarks. But nothing about estimands.



[22] Stevens et al. "A template for the authoring of statistical analysis plans." Contemporary Clinical Trials Communications 34 (2023): 101100.

[23] Gamble et al. "Guidelines for the content of statistical analysis plans in clinical trials." JAMA 318.23 (2017): 2337-2343.

[24] *Fundamental Concepts for New Clinical Trialists*, Scott Evans & Naitee Ting (2015)

SAP examples

- A few I know well:
  - Nielsen et al (2024): https://cdn.clinicaltrials.gov/large-docs/25/NCT04563325/SAP_000.pdf
  - Mortensen et al (2023): https://cdn.clinicaltrials.gov/large-docs/40/NCT03396640/SAP_000.pdf
  - TRACTION trial (2024): https://cdn.clinicaltrials.gov/large-docs/62/NCT06101862/SAP_000.pdf
  - "How long" study (2025): https://cdn.clinicaltrials.gov/large-docs/64/NCT04637464/SAP_000.pdf
  - DURATION RCT: https://cdn.clinicaltrials.gov/large-docs/01/NCT05329701/SAP_000.pdf
  - HALT RCT (2026): https://cdn.clinicaltrials.gov/large-docs/66/NCT05376566/SAP_000.pdf
- Another from an important and recent scandinavian study: (See pages 216–241)

  https://www.nejm.org/doi/suppl/10.1056/NEJMoa2505985/suppl_file/nejmoa2505985_protocol.pdf
- For more examples, just browse supplementary material of papers published in your favorite journals (or "prestigious" journals).

Exercise 4.5 (if time allows): continue reading the SAP for the HALT RCT that you started reading for Exercise 4.4. In preparation for the plenary discussion, select one or a few aspects of the SAP that you would like to discuss. For example, you might raise questions about unclear sections, parts that seem unnecessary or missing, whether the level of detail is appropriate, or whether you could adopt a similar approach in your own study.

